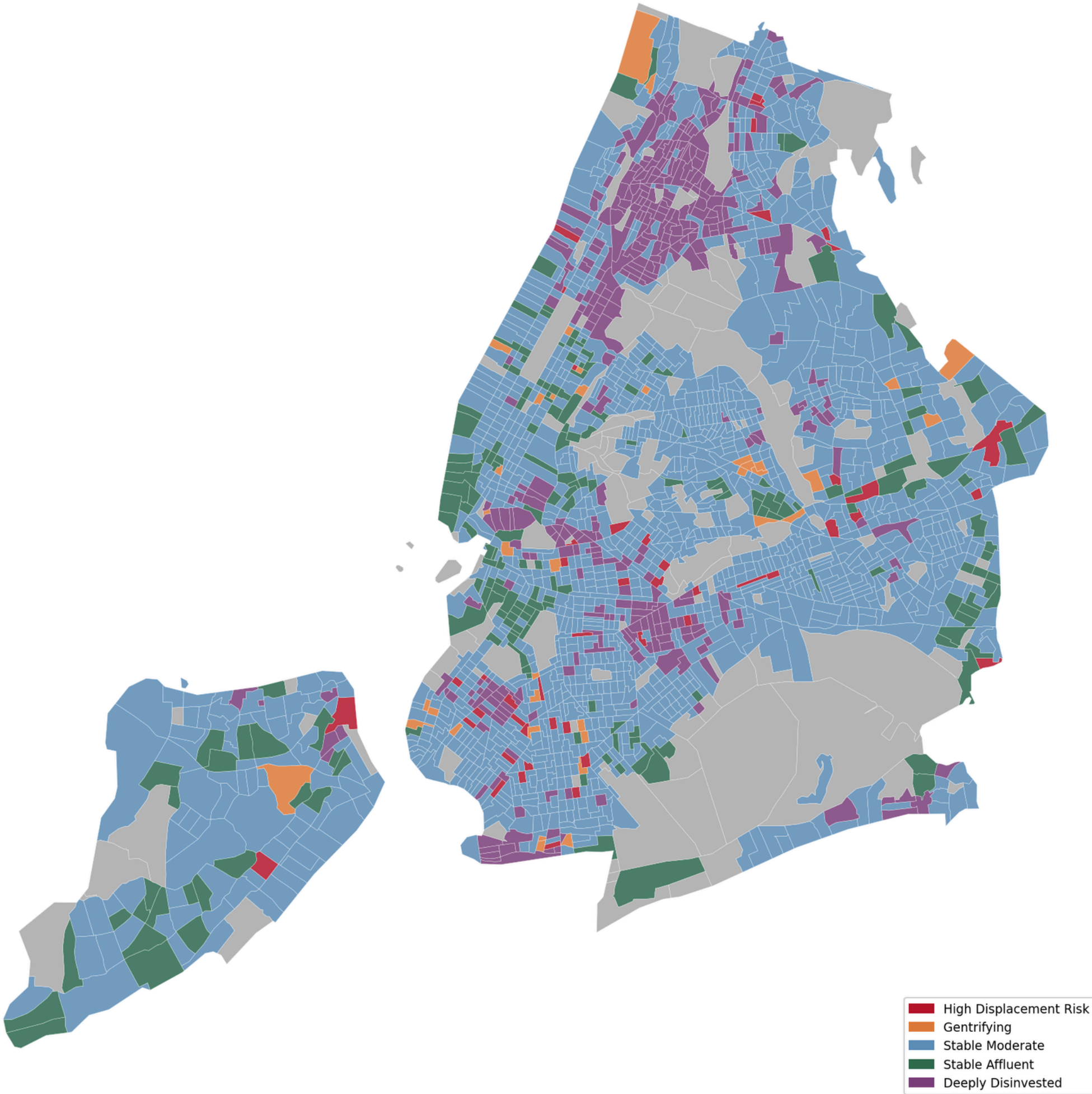


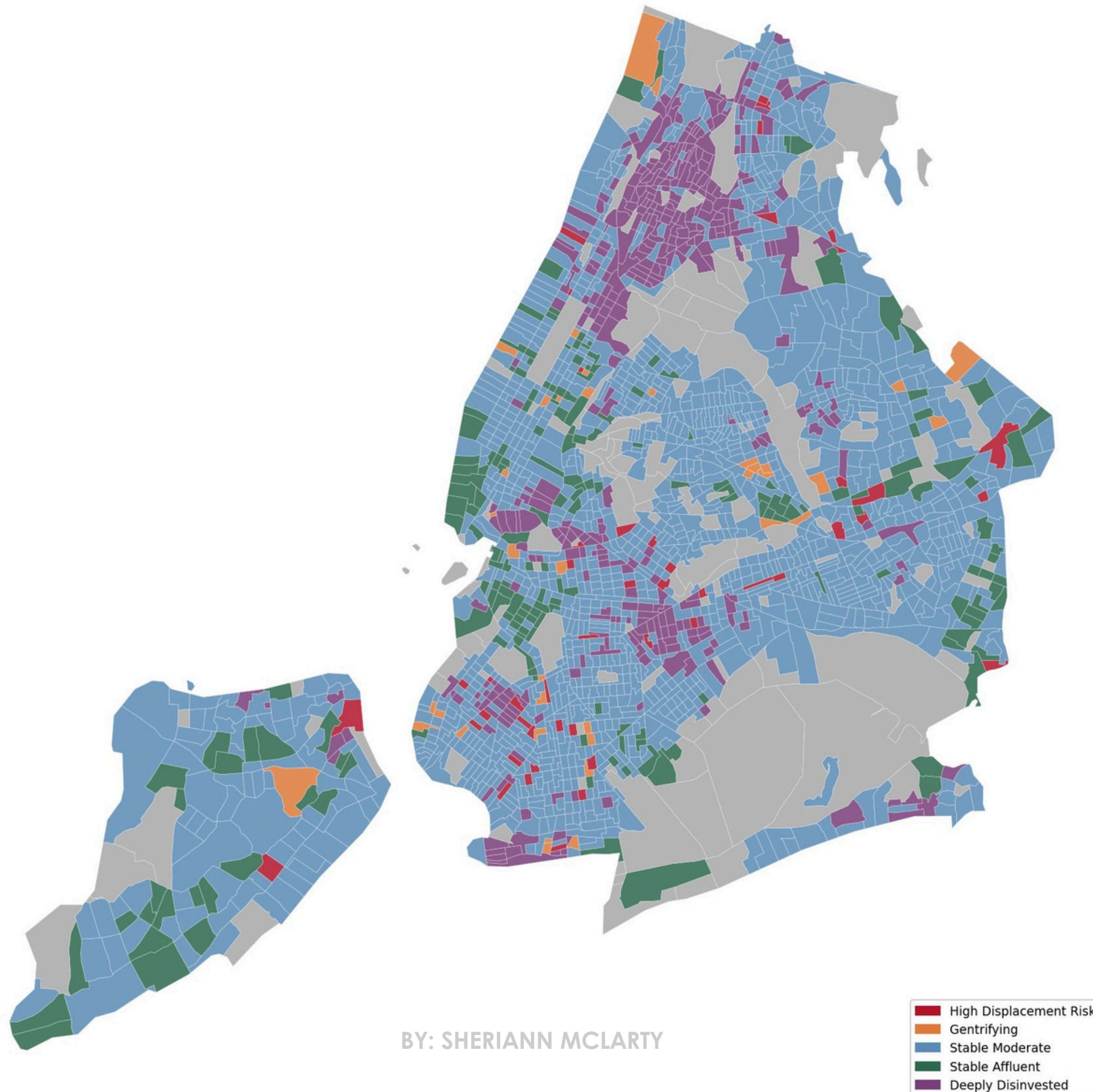
Commuting Networks & Displacement

Gentrification and Network Centrality Across NYC's Five Boroughs

- 5-Borough Commuting Network (LODES 2021)
- ACS Gentrification Classification
- NYC 311 Complaints + NYCHA + PLUTO

Gentrification Stage by Census Tract
NYC Five Boroughs (ACS 2021)





For many first-generation New Yorkers, gentrification means watching your childhood neighborhood change so much that you can no longer afford to live there — and knowing that if something happens to an aging parent who is still there, you cannot afford to move back.

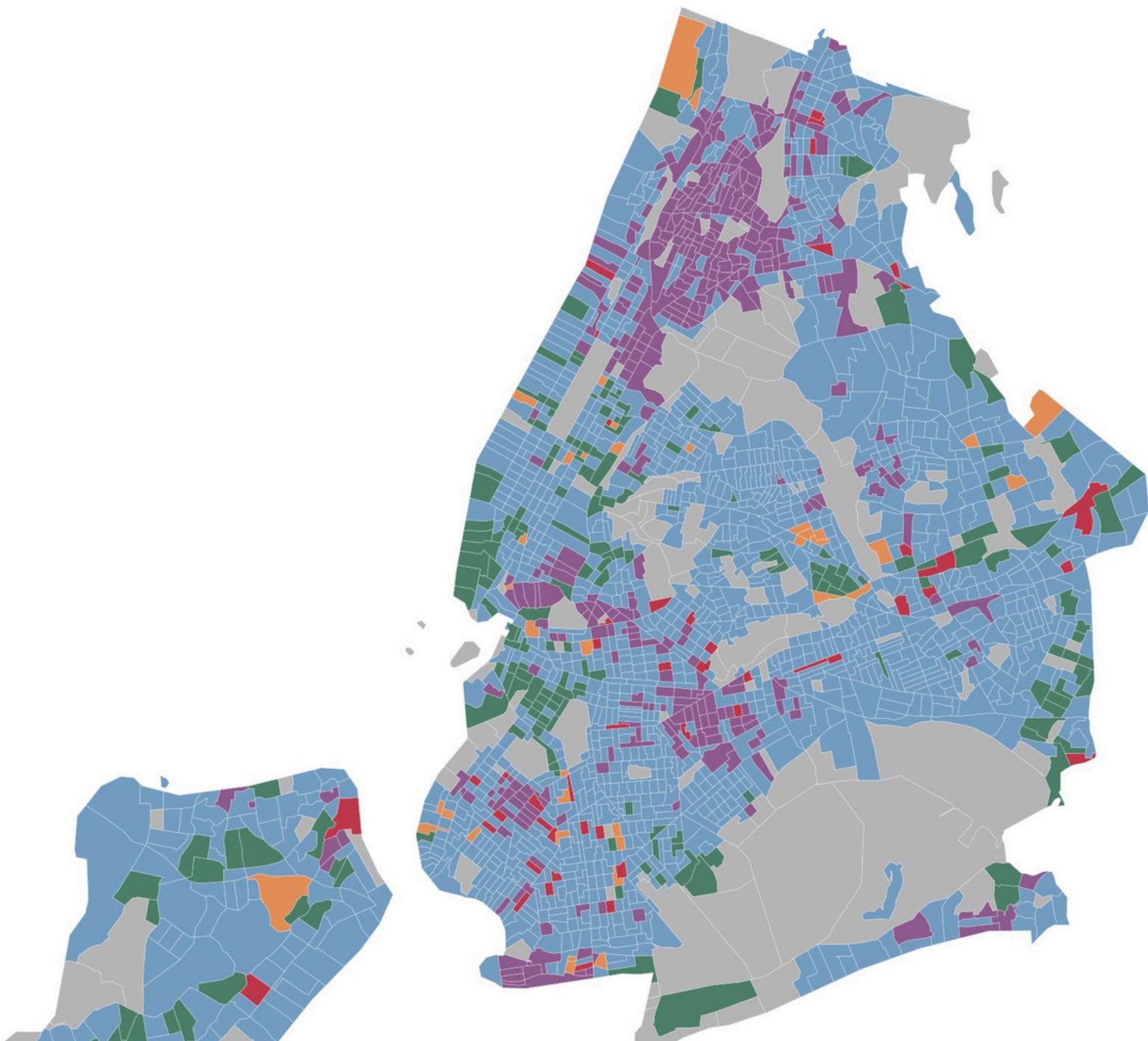
Research Q: Do gentrifying tracts occupy different positions in the NYC commuting network than displaced or disinvested tracts?

Hypothesis: Gentrifying tracts gain in-degree (job hubs). Displaced tracts show higher out-degree — residents commuting farther to find work.

Extends: Project 1 found out-degree was the only significant measure in Brooklyn. This tests whether the pattern holds citywide.

Gentrification Stage Across NYC

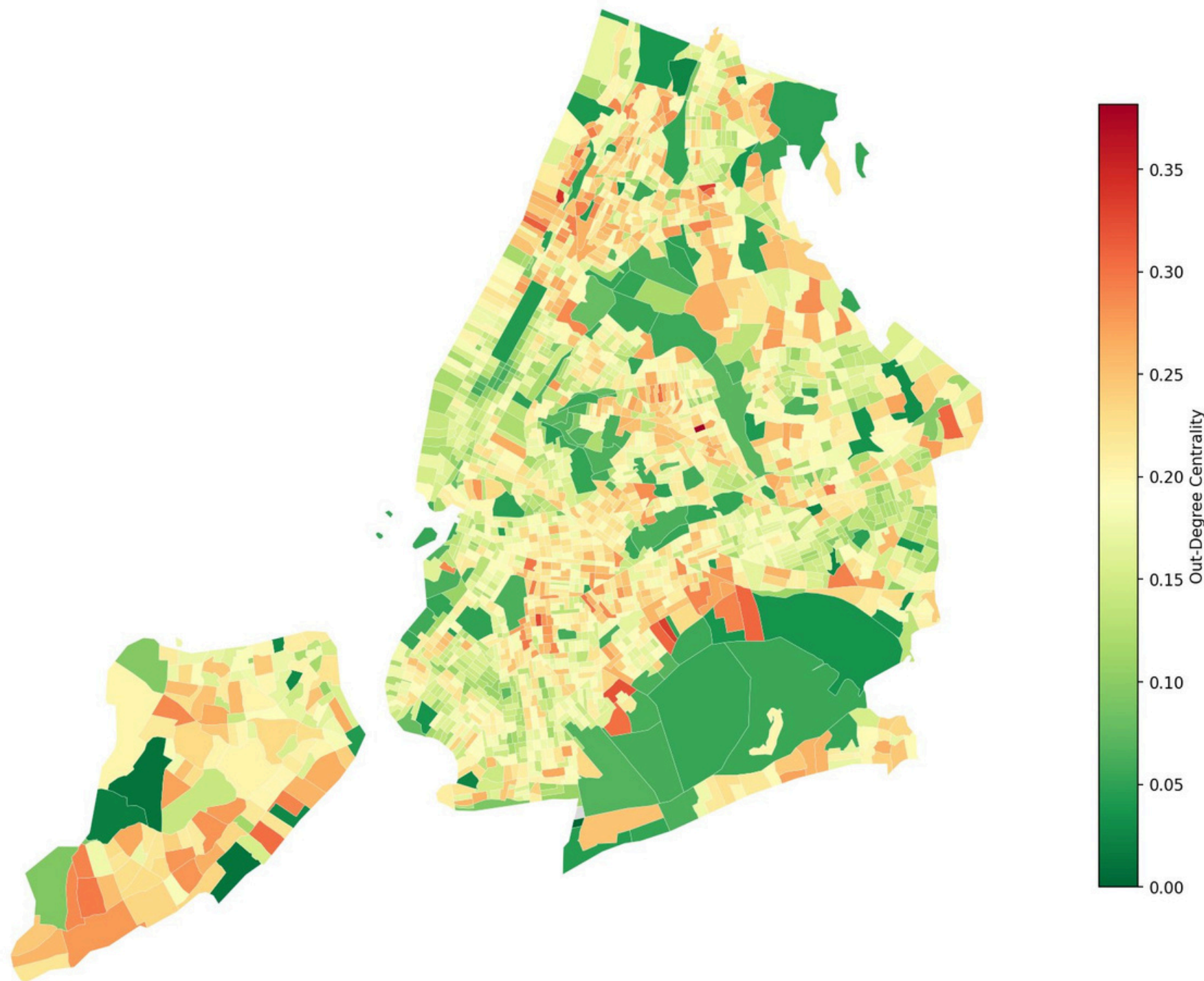
Gentrification Stage by Census Tract
NYC Five Boroughs (ACS 2021)



- High Displacement Risk**
High poverty + high rent burden simultaneously
- Gentrifying**
Rising education + rent burden, mid-flip
- Stable Moderate**
Working class outer-borough stability
- Stable Affluent**
Gentrification already complete
- Deeply Disinvested**
Highest poverty, lowest income — left behind

Where Are Residents Commuting From?

Out-Degree Centrality by Census Tract
NYC Five Boroughs (LODES 2021)



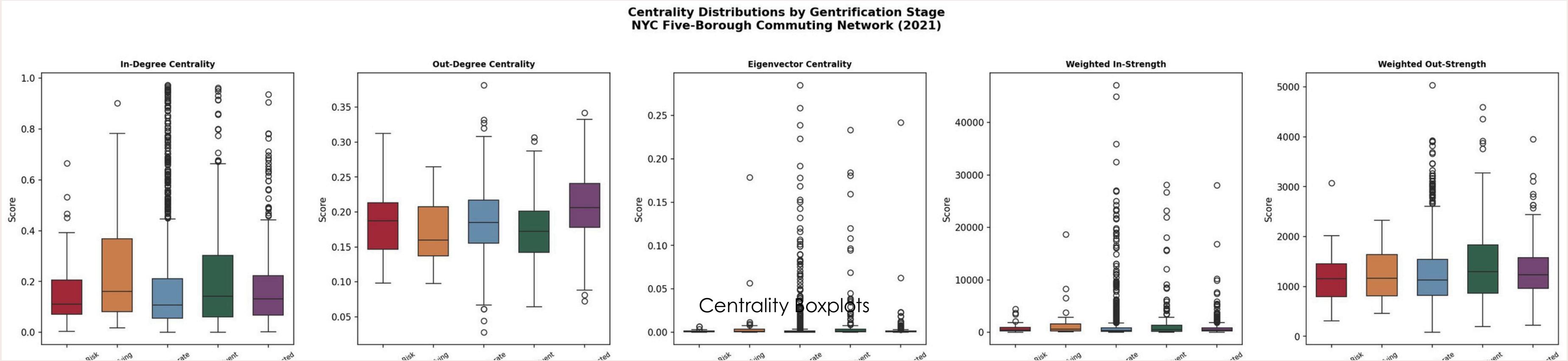
What this shows: Red = high out-degree. Residents commute to many different destinations. Green = low out-degree. Residents stay more local.

The pattern: Red clusters in the South Bronx and central Brooklyn — the same neighborhoods that appear purple (Deeply Disinvested) on the stage map.

Two maps: Overlay these mentally. The same areas that were disinvested are the same areas where residents are forced to commute farthest.

This is not random. It is the predictable result of decades of missing local investment.

Centrality Distributions by Gentrification Stage



In-Degree

Not significant

No clear hub vs peripheral pattern by stage

Out-Degree

Significant ✓

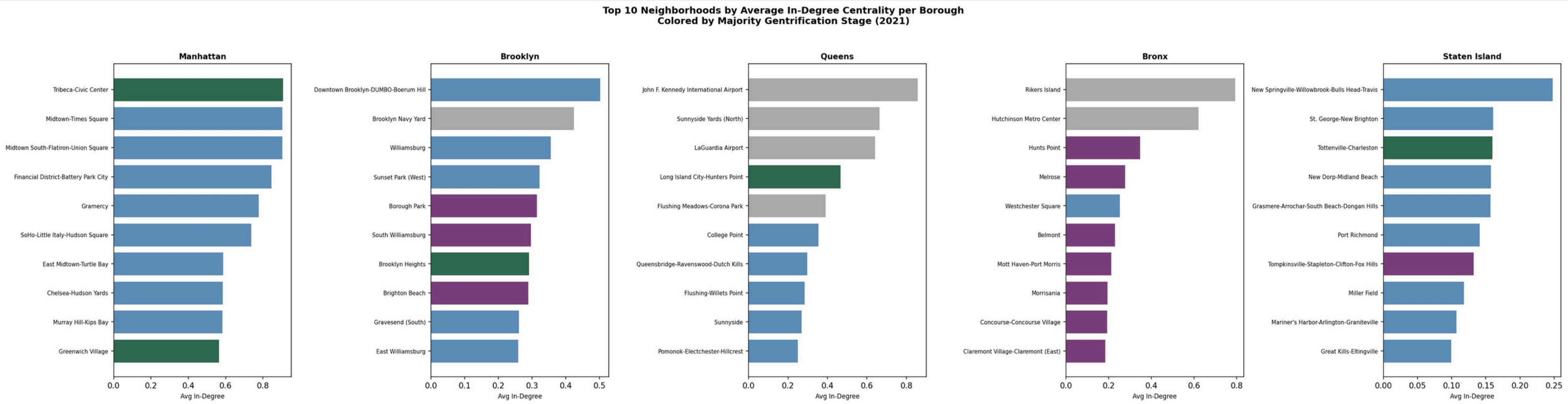
Deeply Disinvested highest median — residents commute most broadly

Eigenvector

Not significant

Embeddedness in commuting core similar across stages

Who Are the Job Hubs by Borough?



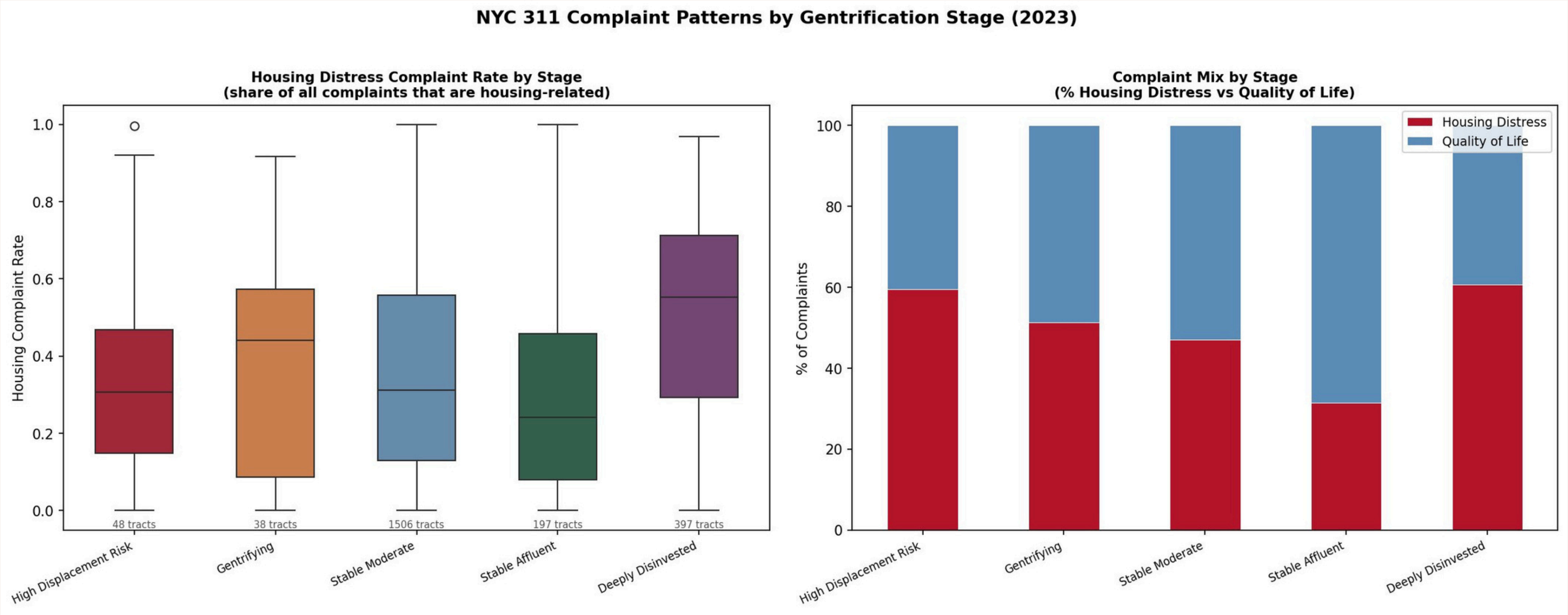
Manhattan: Tribeca, Midtown, Financial District dominate — all Stable Affluent. The job centers are the already-wealthy neighborhoods.

Brooklyn: Downtown-DUMBO tops the list but Borough Park (High Displacement Risk) and South Williamsburg appear in top 6. Hubs are mixed.

Bronx: Rikers Island is the #1 in-degree neighborhood. The top commuting destination in the Bronx is a jail.

Queens: JFK Airport leads — a major job destination for essential workers commuting from across the city.

What Are Residents Complaining About?

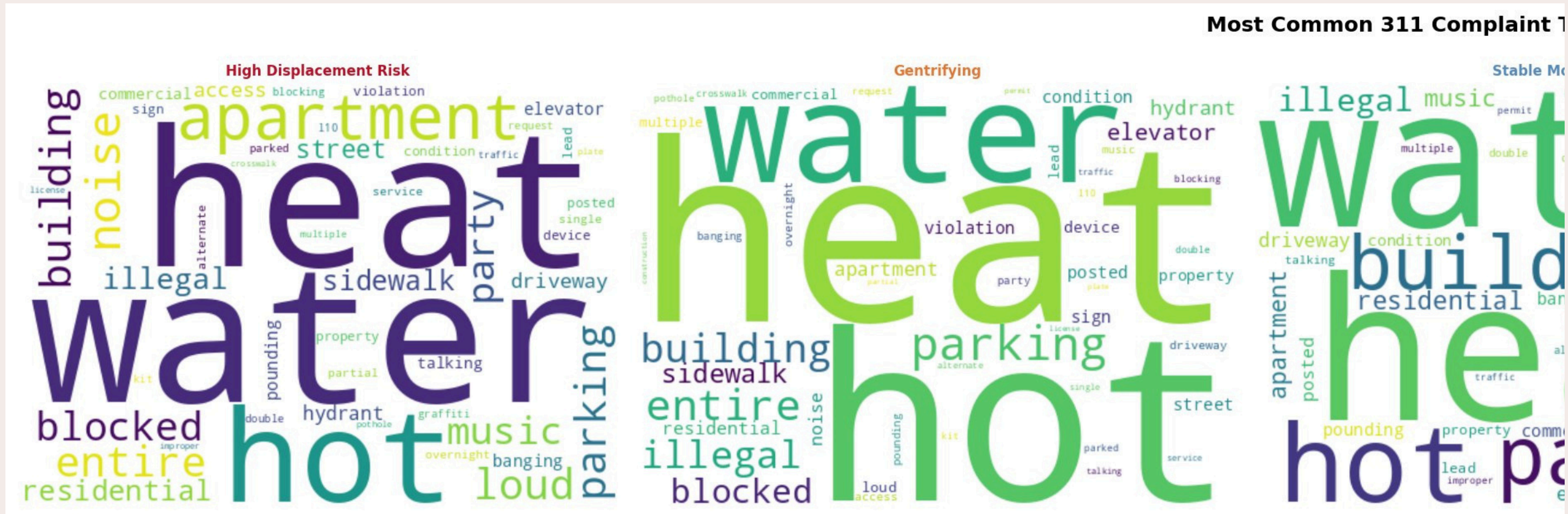


The gap: Two groups of New Yorkers calling the same number about completely different realities.

Deeply Disinvested: 60% of complaints are housing distress — heat, hot water, mold. Pure infrastructure failure.

Stable Affluent: Only 30% housing distress. The rest is quality of life — noise, parking. Not survival complaints. Parking.

The Language of Each Neighborhood



Every stage: heat, water, hot — housing complaints are universal. But look closer.

- **High Displacement Risk:** apartment, illegal, noise — housing conditions AND neighborhood stress.
- **Stable Affluent:** parking, blocked — car-related quality of life. Not survival.
- **Deeply Disinvested:** building, water, heat — pure infrastructure. No parking because that's not the worry.

The Triple Burden

Three compounding disadvantages — measured, not assumed:

01

Worst Transit Access

Deeply Disinvested and High Displacement Risk tracts are farthest from subway stations. Transit deserts are not accidents — they are policy choices.

02

Fastest Rising Rents

2015→2021 ACS data shows rent increases hit hardest in tracts where residents can least afford them. The math does not work.

03

Longest Commutes

High out-degree tracts also have the highest share of workers spending 30+ minutes commuting. Broader travel range = more time lost. Time is money at \$15/hour.

What Do the Purple Tracts Actually Look Like?

NYCHA + PLUTO Data

Public Housing Concentration

Deeply Disinvested tracts hold a disproportionate share of NYCHA developments. The city's housing strategy there was public housing — then systematic underfunding.

Oldest Building Stock

PLUTO data shows these tracts have the highest share of pre-1950 buildings. That means lead paint, poor insulation, aging pipes — exactly what the 311 heat and water complaints are about.

Lowest Assessed Values

Lower assessed values = smaller property tax base = fewer local services. The disinvestment is self-reinforcing. Less money in means less money available to spend.

The network picked up the signal

When economic opportunity was never built locally, residents have no choice but to commute broadly. High out-degree is the predictable result of deliberate disinvestment made visible through data.

Conclusion

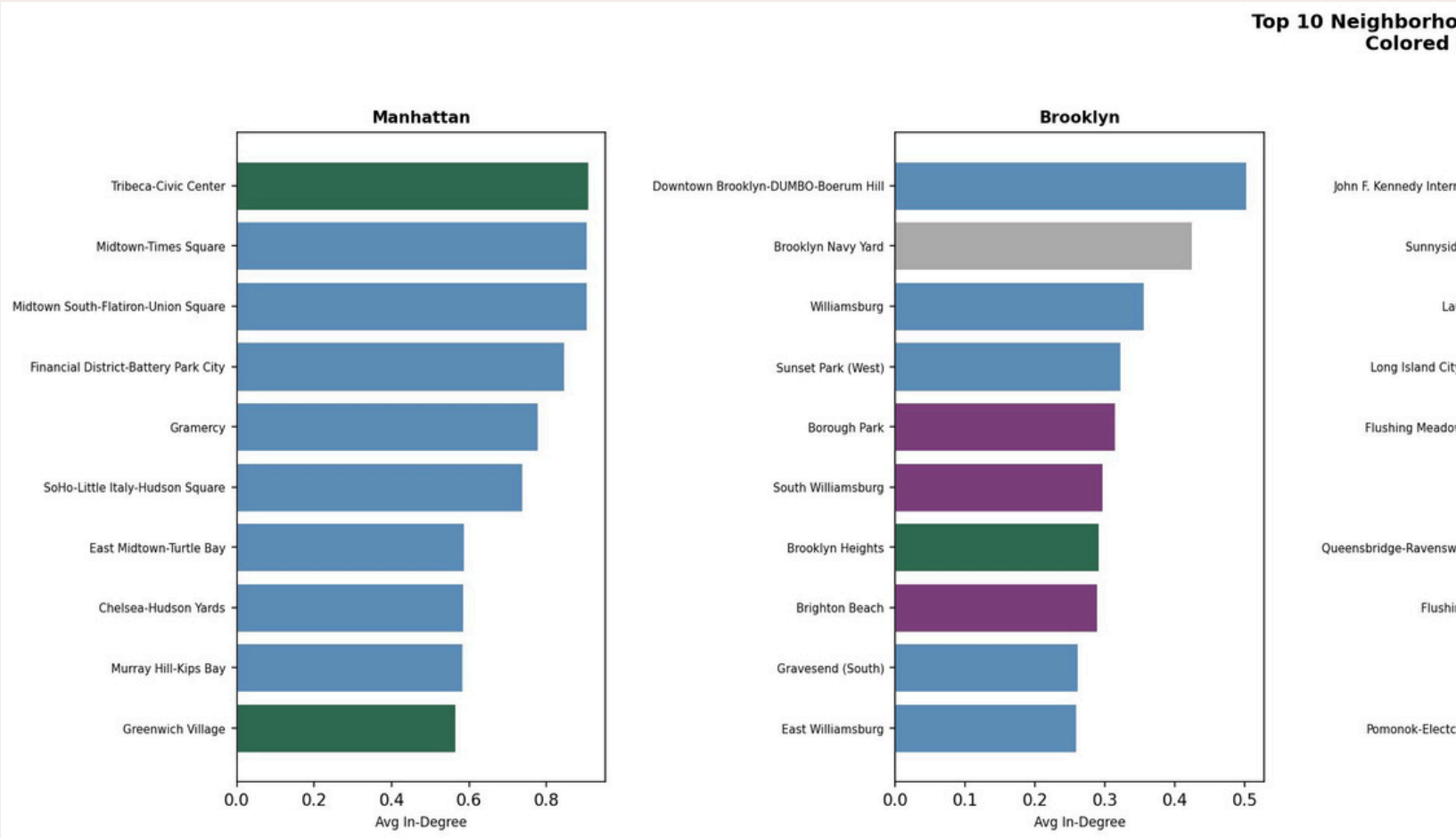
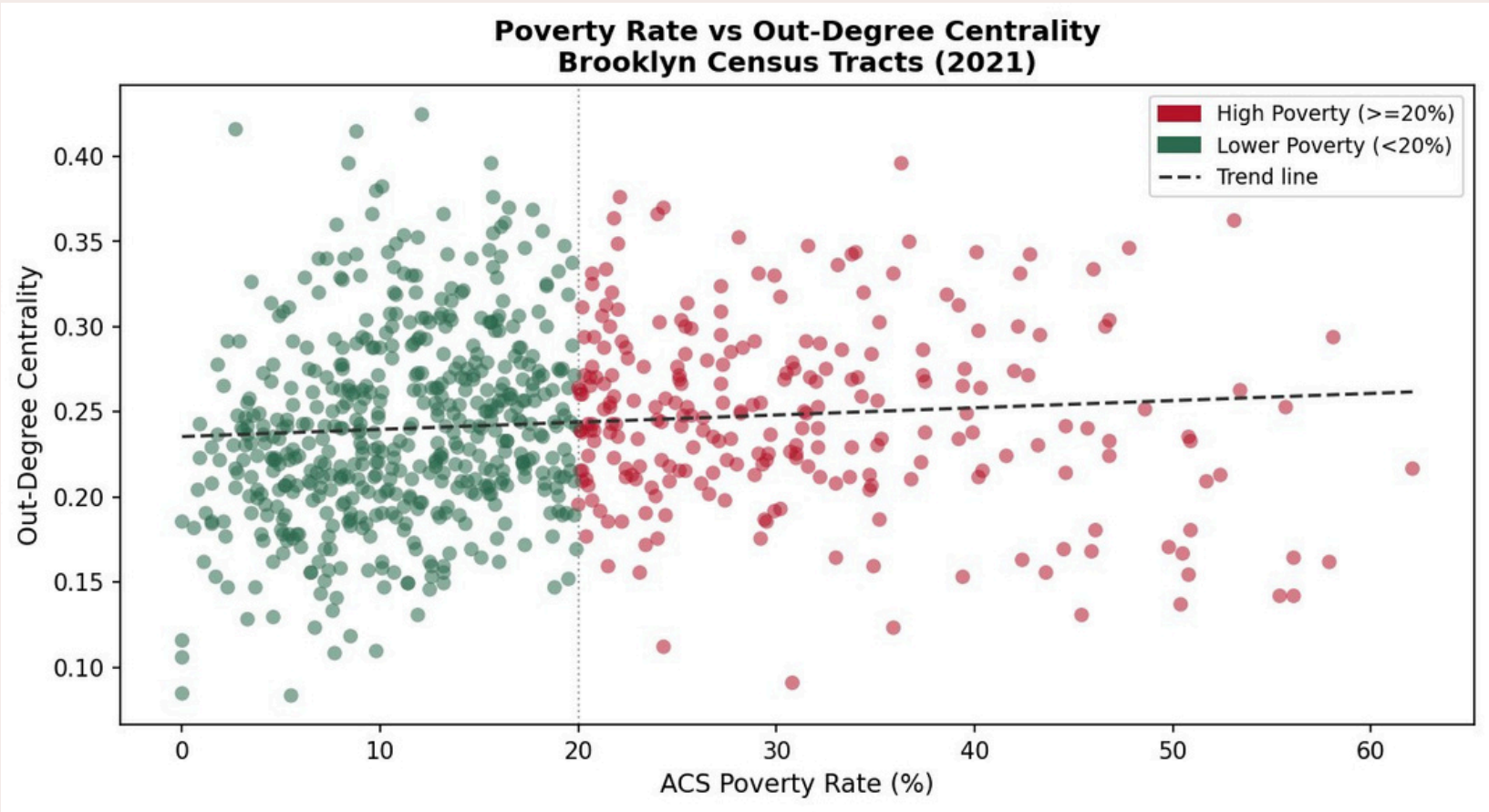
Do gentrification and displacement show up in the commuting network?

In-degree & EVC: Not significant. Hub-peripheral divide does not map cleanly onto gentrification stage.

Out-degree: Significant. Deeply Disinvested tracts have residents commuting to the broadest spread of destinations — not disconnected, dispersed.

311 confirms it: Deeply Disinvested tracts file 60% housing distress complaints. Stable Affluent file 30%. Two cities, one 311 number.

The argument: High out-degree is not a sign of opportunity. It is the fingerprint of forced dispersal — made visible through network data.



Thank You

github.com/sherianmclarty/DATA620_WebAnalytics_SheriannMc

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